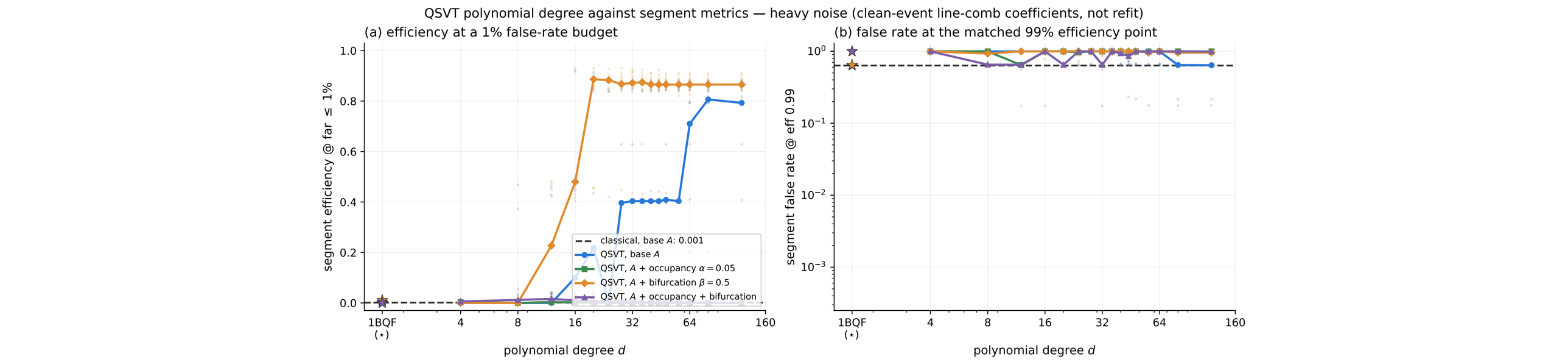

$T = 200$, 10 reps · $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu$m, drop 1% · formula $\varepsilon = 6.31$ mrad · $\gamma = 3$, $\delta = 1$, step kernel · QSVT coefficients: least-squares fit per Hamiltonian in Chebyshev-moment space, μ by eff@far $\leq 1\%$ · every solver at its efficiency-first working point · clean regime omitted: every Hamiltonian reads 1.000 (median of 10 reps) at every degree ≥ 4